

Chapter 5

Systems of Linear Equations: Structure and Strategy

A single equation describes one relationship. Many real problems contain two unknown quantities and two relationships that must be true at the same time. This leads to a *system of equations*.

The main goal of this chapter is not to memorize two procedures. It is to learn how to choose a useful representation, remove an unknown strategically, and interpret what the equations are saying. We shall use substitution and elimination, but we shall also ask why these methods work and when one method is better than the other.

Opening Challenge

A school club sells student tickets for \$4 and adult tickets for \$7. A total of 85 tickets are sold for \$430. How many of each kind are sold?

Let s be the number of student tickets and a the number of adult tickets. Two facts must hold:

$$s + a = 85, \quad 4s + 7a = 430.$$

The first equation counts tickets; the second counts dollars. Neither equation alone determines the answer. Together they do.

From $s = 85 - a$, substitute into the second equation:

$$4(85 - a) + 7a = 430.$$

Thus $340 + 3a = 430$, so $a = 30$ and $s = 55$.

The important lesson is not the numbers 55 and 30. It is the modeling idea: *two unknowns require enough independent relationships to determine them.*

5.1 What Is a System?

A system of two linear equations has the form

$$\begin{aligned}a_1x + b_1y &= c_1, \\a_2x + b_2y &= c_2.\end{aligned}$$

A solution is an ordered pair (x, y) that satisfies *both* equations.

Example 5.1. Check whether $(3, 2)$ solves

$$x + y = 5, \quad 2x - y = 4.$$

Solution. Substitution gives

$$3 + 2 = 5, \quad 2(3) - 2 = 4.$$

Both statements are true, so $(3, 2)$ is a solution.

Professor Shen's Remark. A solution to a system is not merely a value that works in one equation. It must make every condition true at the same time. This habit of checking several conditions at once appears throughout mathematics and science.

5.2 Substitution: Replace One Unknown

Substitution is especially natural when one equation already gives one variable in terms of the other.

Example 5.2. Solve

$$\begin{aligned}x - y &= 4, \\2x + 3y &= 13.\end{aligned}$$

Solution. From the first equation,

$$x = y + 4.$$

Substitute this expression for x into the second equation:

$$2(y + 4) + 3y = 13.$$

Hence

$$5y + 8 = 13, \quad y = 1.$$

Then $x = 5$. Therefore

$$(x, y) = (5, 1).$$

Why does substitution work? If $x = y + 4$, then every occurrence of x may be replaced by the equal quantity $y + 4$. We have not changed the relationship; we have only written it in a more useful form.

Ask a question. If neither equation has a variable with coefficient 1 or -1 , is substitution still the best first choice?

5.3 Elimination: Build an Equation with One Unknown

Sometimes it is easier to combine equations so that one variable disappears. This is the elimination method.

Example 5.3. Solve

$$\begin{aligned}2x + 3y &= 17, \\2x - y &= 5.\end{aligned}$$

Solution. Subtract the second equation from the first:

$$(2x + 3y) - (2x - y) = 17 - 5.$$

Thus

$$4y = 12, \quad y = 3.$$

Using $2x - y = 5$ gives

$$2x - 3 = 5, \quad x = 4.$$

Therefore $(x, y) = (4, 3)$.

Elimination is not a trick. If two equalities are true, then adding or subtracting corresponding sides produces another true equality.

Example 5.4. Solve

$$\begin{aligned}3x + 2y &= 16, \\5x - 3y &= 7.\end{aligned}$$

Solution. No variable cancels immediately. Multiply the first equation by 3 and the second by 2:

$$9x + 6y = 48, \quad 10x - 6y = 14.$$

Add the equations:

$$19x = 62, \quad x = \frac{62}{19}.$$

Then

$$3\left(\frac{62}{19}\right) + 2y = 16,$$

so

$$2y = \frac{118}{19}, \quad y = \frac{59}{19}.$$

Thus

$$(x, y) = \left(\frac{62}{19}, \frac{59}{19}\right).$$

A fractional answer is not a sign that the method failed. The structure of the system determines the answer, not our preference for whole numbers.

5.4 Choose the Method Before Calculating

Good problem solvers do not begin by manipulating symbols at random. They first inspect the equations.

1. If a variable is already isolated, substitution is often fastest.
2. If equal or opposite coefficients already appear, elimination is often fastest.
3. If small multiples can create opposite coefficients, elimination is usually efficient.
4. If the problem asks only for a combination such as $x + y$ or $2x - y$, it may be unnecessary to find x and y separately.

Example 5.5. Suppose

$$3x + 5y = 29, \quad 7x - 5y = 11.$$

Find x .

Solution. Add the equations immediately:

$$10x = 40,$$

so $x = 4$. There is no reason to find y because the question did not ask for it.

Example 5.6. Suppose

$$2x + 3y = 18, \quad 5x + 7y = 43.$$

Find $3x + 4y$ without first solving for both variables.

Solution. We try to build the desired expression from the two left sides. Subtract the first equation from the second:

$$3x + 4y = 25.$$

The requested quantity appears immediately.

Discovery. This is a powerful idea: instead of asking “What are x and y ?”, ask “What combination of the given equations produces the quantity I need?”

5.5 From a Story to Two Equations

When a problem contains two unknowns, name them clearly and look for two independent relationships.

Example 5.7. A jar contains nickels and dimes. There are 38 coins worth \$2.65. How many nickels and how many dimes are there?

Solution. Let n be the number of nickels and d the number of dimes. Counting coins gives

$$n + d = 38.$$

Counting cents gives

$$5n + 10d = 265.$$

Divide the second equation by 5:

$$n + 2d = 53.$$

Subtract $n + d = 38$ to obtain $d = 15$. Therefore $n = 23$.

Ask a new question. Suppose there are N coins, all nickels and dimes, worth C cents. Can you derive formulas for the numbers of nickels and dimes? What conditions must N and C satisfy for the answers to be nonnegative whole numbers?

5.6 One Solution, No Solution, or Infinitely Many?

A system does not always have exactly one solution.

Example 5.8. Consider

$$x + 2y = 6, \quad 2x + 4y = 12.$$

The second equation is exactly twice the first. Every solution of one equation is a solution of the other. There are infinitely many solutions.

Example 5.9. Consider

$$x + 2y = 6, \quad 2x + 4y = 15.$$

If the first equation is doubled, we obtain $2x + 4y = 12$, which contradicts the second equation. Therefore there is no solution.

Thus a system of two linear equations can have one solution, no solution, or infinitely many solutions.

Can You Generalize? Consider

$$\begin{aligned} ax + by &= c, \\ kax + kby &= d. \end{aligned}$$

What relationship between c , d , and k gives infinitely many solutions? What relationship gives no solution?

5.7 Deriving a General Formula

For the system

$$\begin{aligned} ax + by &= e, \\ cx + dy &= f, \end{aligned}$$

we can eliminate y without using any particular numbers. Multiply the first equation by d and the second by b :

$$adx + bdy = de, \quad bcx + bdy = bf.$$

Subtract:

$$(ad - bc)x = de - bf.$$

If $ad - bc \neq 0$, then

$$x = \frac{de - bf}{ad - bc}.$$

Similarly,

$$y = \frac{af - ce}{ad - bc}.$$

The quantity $ad - bc$ tells us something important. When it is nonzero, the system has exactly one solution. When it is zero, we must examine the system more carefully: the equations may describe the same relationship or incompatible relationships.

Do not memorize these formulas for routine problems. The important achievement is understanding how they were *derived* by elimination.

5.8 How to Learn from a Solved System

After solving a system, train yourself to ask:

1. Could I have chosen a faster method?
2. What did each equation represent?
3. Could I have found the requested quantity without solving for every unknown?
4. What happens if one constant is changed?
5. Can I replace the numbers by letters and derive a general result?
6. Can I create a different story that leads to the same system?

These questions change algebra from a collection of procedures into a way of thinking.

What Have We Discovered?

1. A solution of a system must satisfy all equations simultaneously.
2. Substitution replaces one quantity by an equal quantity.
3. Elimination creates a new equation in fewer unknowns by combining true equalities.
4. The best method depends on the structure of the equations.
5. Sometimes the desired combination can be found without solving the full system.
6. Word problems require enough independent relationships to determine the unknowns.
7. A linear system may have one solution, no solution, or infinitely many solutions.
8. General formulas should be derived from the basic methods rather than memorized without explanation.

Do not ask only, “How do I solve it?” Ask, “What structure can I use?”

Exercises

Problems 1–10 are Core Exercises and contain the essential work for this chapter. Problems 11–20 are Challenge Problems at approximately AMC 8 and MATHCOUNTS level. They are optional for students whose main goal is a solid mathematical foundation, but are strongly recommended for students preparing for competitions. All exercises below have been newly written for this book.

Part A. Core Exercises

1. Solve

$$\begin{aligned}x + y &= 11, \\x - y &= 3.\end{aligned}$$

2. Solve

$$\begin{aligned}x &= 2y + 1, \\3x + y &= 17.\end{aligned}$$

3. Solve

$$\begin{aligned}2x + 3y &= 19, \\2x - y &= 7.\end{aligned}$$

4. Solve

$$\begin{aligned}3x + 4y &= 18, \\5x - 2y &= 4.\end{aligned}$$

5. Suppose

$$4x + 3y = 26, \quad 4x - 3y = 14.$$

Find x without finding y first.

6. Suppose

$$2x + 5y = 31, \quad 7x + 9y = 74.$$

Find $5x + 4y$ without first solving for x and y separately.

7. A theater sells 64 tickets. Student tickets cost \$5 and adult tickets cost \$8. The total revenue is \$398. How many student tickets and adult tickets are sold?
8. A farm has chickens and goats. There are 29 animals and 82 legs. How many chickens and how many goats are there?
9. Decide whether the system has one solution, no solution, or infinitely many solutions:

$$2x + 6y = 10, \quad x + 3y = 5.$$

10. Find the value of k for which the system

$$x + 2y = 7, \quad 3x + 6y = k$$

has infinitely many solutions. For what values of k does it have no solution?

Part B. Challenge Problems

11. Positive numbers
- x
- and
- y
- satisfy

$$x + y = 24, \quad \frac{x}{3} + \frac{y}{4} = 7.$$

Find $x - y$.

12. A collection of 50 coins consists only of quarters and dimes and has a total value of \$8.60. How many quarters are there?

13. Two numbers have sum 41. If twice the larger number is 7 more than three times the smaller number, find the product of the two numbers.

14. Suppose

$$3x + 2y = 17, \quad 5x - 4y = 1.$$

Find $8x - 2y$ without first finding x and y separately.

15. The system

$$(k - 1)x + 2y = 6, \quad 4x + (k + 1)y = 9$$

has exactly one solution for most values of k . Find all values of k for which it does not have exactly one solution.

16. A two-digit number has digit sum 11. Reversing its digits decreases the number by 27. Find the number.

17. At a school event, 120 sandwiches are sold. A regular sandwich costs \$4 and a large sandwich costs \$6. After all sandwiches are sold, the money collected is \$588. Later it is discovered that exactly 12 customers who bought large sandwiches were accidentally charged the regular price. How many large sandwiches were actually sold?

18. Three times one number plus twice another is 50. If 4 is added to the first number and 1 is subtracted from the second, the two new numbers are equal. Find the original two numbers.

19. The equations

$$ax + by = 18, \quad bx + ay = 12$$

have the solution $x = 4$, $y = 2$. Find $a + b$ and $a - b$.

20. A student knows that

$$2x + 3y = 19$$

and wants to determine x and y uniquely by adding one more linear equation. Which of the following would *not* be enough, and why?

- (A) $x - y = 1$, (B) $4x + 6y = 38$,
 (C) $3x + 2y = 16$, (D) $y = 3$.

Part C. Beyond the Chapter

These questions are optional. Their purpose is to practice asking questions, deriving formulas, and summarizing ideas.

21. Starting from

$$ax + by = e, \quad cx + dy = f,$$

derive formulas for x and y by elimination. Explain in words why the quantity $ad - bc$ is important.

22. A store sells two products at prices p and q dollars. A total of N items are sold for R dollars. Derive formulas for the number of each product sold. What restrictions must the data satisfy if the answers represent actual numbers of items?
23. Create two different systems that both have $(x, y) = (3, -2)$ as their unique solution. Then create one system with no solution and one with infinitely many solutions. Explain how you designed each system.
24. Write a six-sentence summary of this chapter. Your summary must explain substitution, elimination, how to choose between them, and the three possible numbers of solutions. End by writing one question about systems of equations that you would like to investigate further.

Answer Key

- | | |
|----------------------------|-------------------------------|
| 1. (7, 4) | 2. (5, 2) |
| 3. (5, 3) | 4. (2, 3) |
| 5. 5 | 6. 43 |
| 7. 38 student, 26 adult | 8. 17 chickens, 12 goats |
| 9. infinitely many | 10. $k = 21$; otherwise none |
| 11. 0 | 12. 24 quarters |
| 13. 390 | 14. 18 |
| 15. $k = -3, 3$ | 16. 74 |
| 17. 66 large | 18. 8, 13 |
| 19. $a + b = 5, a - b = 3$ | 20. (B) |

For Problem 20, equation (B) is just twice the original equation, so it adds no new independent information.

For Problem 21, when $ad - bc \neq 0$,

$$x = \frac{de - bf}{ad - bc}, \quad y = \frac{af - ce}{ad - bc}.$$

Problems 22–24 include open-ended explanation and creation. A strong response should clearly identify independent relationships, justify any restrictions, and explain the reasoning rather than merely state an answer.